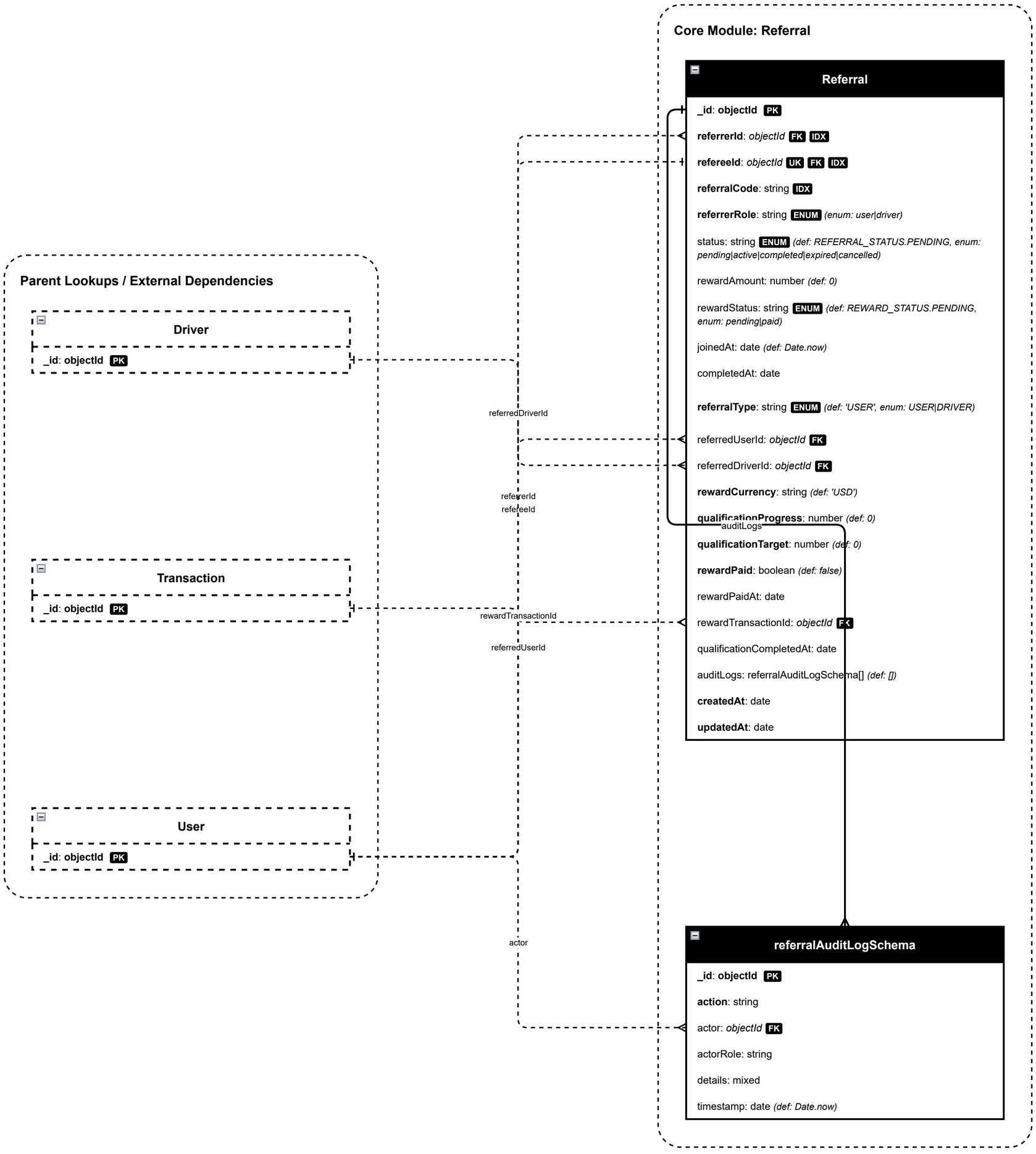
Core Module: Referral
Referral
_id: objectId PK
referrerId: objectId FK IDX
refereeId: objectId UK FK IDX
referralCode: string IDX
referrerRole: string ENUM (enum: user|driver)
status: string ENUM (def: REFERRAL_STATUS.PENDING, enum: pending|active|completed|expired|cancelled)
rewardAmount: number (def: 0)
rewardStatus: string ENUM (def: REWARD_STATUS.PENDING, enum: pending|paid)
joinedAt: date (def: Date.now)
completedAt: date
referralType: string ENUM (def: 'USER', enum: USER|DRIVER)
referredUserId: objectId FK
referredDriverId: objectId FK
rewardCurrency: string (def: 'USD')
qualificationProgress: number (def: 0)
qualificationTarget: number (def: 0)
rewardPaid: boolean (def: false)
rewardPaidAt: date
rewardTransactionId: objectId FK
qualificationCompletedAt: date
auditLogs: referralAuditLogSchema[] (def: [])
createdAt: date
updatedAt: date
auditLogs
referralAuditLogSchema
_id: objectId PK
action: string
actor: objectId FK
actorRole: string
details: mixed
timestamp: date (def: Date.now)
Parent Lookups / External Dependencies
Driver
_id: objectId PK
Transaction
_id: objectId PK
User
_id: objectId PK
referredDriverId
referrerId
refereeId
rewardTransactionId
referredUserId
actor